

MD OMOR FARUK

Ph.D. Student, Graduate Research Assistant
Department of Electrical Engineering and Computer Science
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CAREER OBJECTIVE

I am planning to develop robust, scalable, and physics-informed reinforcement learning algorithms for large-scale multi-agent systems aimed at addressing real-world applications while significantly reducing the computational complexity of state-of-the-art algorithms. In addition, I seek to enhance my expertise in solving innovative problems, designing sustainable and novel technologies, and advancing the frontiers of Artificial Intelligence, 3D reconstruction, Reinforcement Learning, and Communication Systems.

EDUCATIONAL QUALIFICATION

College of Engineering and Physical Sciences, University of Wyoming, Laramie, WY, USA

Doctor of Philosophy (Ph.D.) Student, Electrical Engineering and Computer Science (*Fall 2023 – Ongoing*)

Faculty of Engineering, University of Rajshahi, Rajshahi, Bangladesh

Master of Engineering (M. Eng.) Applied Physics and Electronic Engineering (Current Department of Electrical and Electronic Engineering), CGPA: 3.33 out of 4.00 (*November 2017*)

Faculty of Engineering, University of Rajshahi, Rajshahi, Bangladesh

Bachelor of Science in Engineering (B.Sc. Eng.) Applied Physics and Electronic Engineering (Current Department of Electrical and Electronic Engineering), CGPA: 3.29 out of 4.00 (*October 2015*)

TECHNICAL SKILLS

- **Programming and Software:** Python, CUDA (NumPy, SciPy, scikit-learn, Matplotlib, OpenAI Gym); Deep Learning Frameworks (PyTorch, JAX, Tensorflow, etc.), Physics Simulation Frameworks (Issac Sim/Lab, MuJoCo, etc.), MATLAB/Simulink (communication and control systems, signal processing); C, C++ (embedded and performance-critical modules)
- **Machine Learning and AI:** Reinforcement Learning (Deep RL, Multi-Agent RL, Mean-Field Games); Neural Networks (CNNs, RNNs, PINNs); Data Analysis and Visualization (Pandas, Seaborn, Matplotlib)
- **Robotics and Control:** Multi-Agent Formation Control; UAV Path Planning and Control (simulation, real-world testbeds: Crazyflie, OptiTrack, CoppeliaSim); ROS (Robot Operating System)
- **Networking Skill (CCNA Exploration):** IP addressing, subnetting, and routing protocols (RIP, OSPF, DHCP), Switching and VLAN configuration, Wireless networking and WAN technologies, Network troubleshooting and performance optimization
- **Web Development (Front-end):** CSS, CSS-advanced, HTML, HTML5
- **Simulation Software:** Xilinx 9.2i, WinAVR, AVR studio 4, Proteus 7.7
- **PCB Designing**

WORK EXPERIENCE

Graduate Research Assistant

Department of Electrical Engineering and Computer Science, University of Wyoming, Laramie, WY, USA,
From Fall 2023 - Fall 2024, Summer 2025 – Continuing

Graduate Teaching Assistant and Grader

Department of Electrical Engineering and Computer Science, University of Wyoming, Laramie, WY, USA,
Spring 2025

Research Assistant

Wireless Communication Research Group, Department of Electrical and Electronic Engineering, University of Rajshahi, Rajshahi 6205, Bangladesh, *From Jan 2017 – July 2023*

Business Analyst

Kazi IT Center Ltd, 5th Floor, Dainik Barta Complex, Alupotti, Rajshahi – 6100, *From July 2017 – May 2019*

TEACHING EXPERIENCE

Instructor

Research Experience for Undergraduates Project, Center for Controlled Environment Agriculture (CEA), University of Wyoming, Laramie, WY 82071, USA, *Summer 2025*

Graduate Teaching Assistant

Department of Electrical Engineering and Computer Science, University of Wyoming, Laramie, WY 82071, USA, *Spring 2025*

PUBLICATIONS

1. Sumona, Rabeya Bashri, John Pritom Biswas, Ahmed Shafkat, Md Mahbubur Rahman, **Md Omor Faruk**, and Yaqoob Majeed. "An integrated deep learning approach for enhancing brain tumor diagnosis." *Healthcare Analytics* (2025): 100421. <https://doi.org/10.1016/j.health.2025.100421>
2. **Faruk, Md Omor**, Saifur Rahman Sabuj, SK Tamanna Kamal, Fowzia Jabin, Joarder Jafor Sadique, and Shaikh Enayet Ullah. "Multi-user non-orthogonal waveform transceiver in RIS-assisted UAV communication: design and implementation." *International Journal of Vehicle Information and Communication Systems* 10, no. 1 (2025): 85-111. <https://doi.org/10.1504/IJVICS.2025.144103>
3. Uddin, Mohammad Irfan, Nishad Tasnim, **Md Omor Faruk**, and Zejian Zhou. "Energy-Efficient Deep Reinforcement Learning with Spiking Transformers." *arXiv preprint arXiv:2505.14533* (2025). <https://doi.org/10.48550/arXiv.2505.14533>
4. Tasnim, Nishad, **Md Omor Faruk**, and Zejian Zhou. "Optimal Dynamic Load Altering Attack on Power Grids via Electrical Vehicle Charging Stations." In *2024 Cyber Awareness and Research Symposium (CARS)*, pp. 1-6. IEEE, 2024. <https://doi.org/10.1109/CARS61786.2024.10778813>
5. **Faruk, Md Omor**, Md Najmul Hossain, Shaikh Enayet Ullah, SK Tamanna Kamal, and Fowzia Jabin. "Transceiver Design for 6G Compatible CP-less Multi-User Orthogonal Chirp Division Multiplexing System with Physical Layer Security." *Authorea Preprints* (2023). <https://doi.org/10.36227/techrxiv.22214362.v1>
6. **Omor Faruk, Md**, Joarder Jafor Sadique, Kanapathippillai Cumanan, and Shaikh Enayet Ullah. "Orthogonal variable spreading factor encoded unmanned aerial vehicle-assisted nonorthogonal multiple access system with hybrid physical layer security." *ETRI Journal* 45, no. 2 (2023): 213-225. <https://doi.org/10.4218/etrij.2021-0292>
7. **Faruk, Md Omor**, Jinia Rahman, and Shaikh Enayet Ullah. "Performance analysis of downlink multiuser CP-free MIMO OFDM system with interference cancellation and enhancement of physical-layer security." *European Journal of Computer Science and Information Technology* 7, no. 3 (2019): 26-40. <https://www.researchgate.net/publication/334317414>
8. Naher, Mst Afrin, **Md Omor Faruk**, Md Mahbubur Rahman, and Shaikh Enayet Ullah. "Performance Evaluation of Encrypted Text Message Transmission in 5G Compatible Frequency-domain Subband Superposed Scheme Implemented MIMO OFDM Wireless Communication System." *Global Journal of Computer Science and Technology* (2019). <https://scispace.com/pdf/performance-evaluation-of-encrypted-text-message-m2ijcn7v01.pdf>
9. Datta, Srabanti, and **Md Omor Faruk**. "Design & performance evaluation of a dipole antenna using various lengths of antenna material." *International Journal of Scientific & Engineering Research* 10, no. 1. <https://www.researchgate.net/publication/331194354>
10. Datta, Srabanti, and **Md Omor Faruk**. "Impact of permittivity and radiation pattern on the performance of a dipole antenna." *Intern. J. Progres. Sci. Technol* 17, no. 1 (2019): 191-196. <https://ijpsat.org/index.php/ijpsat/article/view/1377>
11. **Faruk, Md. Omor**, and Shaikh Enayet Ullah. "Encrypted Color Image Transmission in LQ-Based GSIC Pre-Coded Multiuser Downlink Wireless Communication System." *International Journal of Advances in Image and Video Processing*, Society for Science and Education 7, no. 4 (2019): 6-15. https://www.researchgate.net/publication/335841104_Encrypted_Color_Image_Transmission_in_LQ-based_GSIC_Pre-coded_Multiuser_Downlink_Wireless_Communication_System

12. **Faruk, Md Omor**, and Shaikh Enayet Ullah. "Performance Evaluation of Encrypted Text Message Transmission in 5G Compatible Orthogonal Multi-level Chaos Shift Keying Modulation Scheme Aided MIMO Wireless Communication System." (2018). <https://www.researchgate.net/publication/330180334>
13. **Faruk, Md Omor**, and Shaikh Enayet Ullah. "Performance evaluation of orthogonal multi-level chaos shift keying modulation scheme aided MIMO wireless communication system." *International Journal of Networks and Communications* 8, no. 1 (2018): 10-17. <http://article.sapub.org/10.5923.j.ijnc.20180801.02.html>

RESEARCH ARTICLES UNDER REVIEW

1. **Md Omor Faruk**, Nishad Tasnim, Mohammad Irfan Uddin, and Zejian Zhou. "A Physics-Informed Reinforcement Learning Framework for Massive Multi-Agent Control." *International Conference on Robotics & Automation (ICRA) 2026, Vienna, Austria*.
2. Uddin, Mohammad Irfan, Nishad Tasnim, **Md Omor Faruk**, and Zejian Zhou. "STRL: Integrating Spiking Neural Dynamics with Transformers for Energy-Efficient Reinforcement Learning." *The 40th Annual AAAI Conference on Artificial Intelligence 2026, Singapore*.
3. Tasnim, Nishad, **Md Omor Faruk**, and Zejian Zhou. "Optimal Dual-Role Attack and Defense for Power Grid Security: A Decision Transformer Approach." *57th North American Power Symposium, Connecticut, USA (Accepted)*.
4. Jahanur Biswas, Md Nahid Hasan, Md. Muksit Ul Islam, Md Mahbubur Rahman, Ali Torabi, Sanjog Gaihre, **Md Omor Faruk**, Yaqoob Majeed, "Performance-Optimized Alzheimer's Detection Using Machine Learning with SMOTE and Randomized Hyperparameter Tuning." *SN Computer Science, Springer*.

RESEARCH JOURNAL/CONFERENCE REVIEWER

- Reviewer of International Journal of Communication Systems, John Wiley and Sons Ltd
- Reviewer of the 40th Annual AAAI Conference on Artificial Intelligence, *Singapore 2026*
- Reviewer of IEEE International Conference on Robotics and Automation (ICRA), *Vienna, Austria 2026*

RESEARCH CREDENTIALS

Google Scholar: <https://scholar.google.com/citations?user=aOR1bMsAAAAJ&hl=en>

AWARDS & SCHOLARSHIPS

- **ASUW Leadership Scholarship**, University of Wyoming, Laramie, WY 82071, USA 2025-2026
- **International Student Scholarships**, University of Wyoming, Laramie, WY 82071, USA 2025-2026
- **GPA 5 Talent Award**, Lahiri Student Welfare Organization, Lahiri, Thakurgaon 5141, Bangladesh 2009

PROFESSIONAL SOCIETY MEMBERSHIP

- Member of IEEE (Institute of Electrical and Electronics Engineers)
- Member of IEEE Student Chapter, University of Wyoming, Laramie, WY 82071, USA
- Member of ACM Student Chapter, University of Wyoming, Laramie, WY 82071, USA

ADDITIONAL COURSE

Course on **Web Development** by Ernst & Young LLP, India under Leveraging ICT for Growth, Employment and Governance (LICT) Project of Bangladesh Computer Council (BCC), ICT Division, People's Republic of Bangladesh. *December 2017*

Developed Comprehensive ideas on

Machine Learning and Artificial Intelligence

Supervised learning, unsupervised learning, support vector machines (SVM), k-nearest neighbors (KNN), random forests, principal component analysis (PCA), long short-term memory (LSTM), feature engineering, model evaluation metrics, bias and variance tradeoff, overfitting and underfitting, hyperparameter tuning.

Neural Networks and Deep Learning

Artificial neurons and perceptron's, multi-layer perceptron's (MLP), activation functions (ReLU, Sigmoid, Tanh, SoftMax), feedforward neural networks, backpropagation and gradient descent, loss functions (cross-entropy, mean squared error), optimization techniques (SGD, Adam), convolutional neural networks (CNN), pooling layers (max pooling, average pooling), recurrent neural networks (RNN), autoencoders, generative adversarial networks (GANs), transformers and attention mechanisms, regularization techniques (dropout, L1/L2 regularization), batch normalization, neural network architectures.

Reinforcement Learning (RL)

Agent-environment interaction, reward signals, Markov decision processes (MDP), exploration vs. exploitation tradeoff, policy-based methods, value-based methods, Q-learning, deep Q-networks (DQN), actor-critic algorithms, Monte Carlo methods, temporal difference (TD) learning, state-value function, action-value function, Bellman equation, discount factor, multi-arm bandit problem, SARSA (State-Action-Reward-State-Action) algorithm, reinforcement learning in robotics, self-play, inverse reinforcement learning, multi-agent reinforcement learning, applications in gaming, and autonomous systems.

Computer networks and data communication

The OSI reference model, TCP/IP reference model, LAN, WAN, Packet switching and Fast packet switching, DNS, IP protocol, IPv4, IPv6, Subnet, Virtual Private Network (VPN), Mobile IP, Network security, SNMP, Router, Switches, DHCP, Wireless LAN protocol, High speed LANs.

Future generation (5G) wireless networking

Frequency spectrum allocation, mmWave Communications for 5G, applicability of hybrid beamforming techniques with higher order digital modulations (256 QAM and 256-PSK) and large-scale planar antenna systems in 5G, Emerging Applications of mmWave Communications (Information Showers, optical-to-RF inter connectivity, V2V, V2I, wireless internet access and wireless connectivity for seat back entertainment in aircraft).

LTE-Advanced based 4G Networking

Basics of LTE, LTE Release 8(3.9G), Carrier aggregation, network architecture of LTE Advanced system, advanced downlink and uplink MIMO technologies, Relaying, Antenna deployment configurations (Closely/widely spaced cross-polarized antenna, Uniform linear array).

WiMAX Networking

OFDM-based physical layering, Scalable bandwidth and data rate support, Mobile WiMAX Network Planning with special emphasis on collecting Geographical, Customer and Technology information and consideration of coverage area with signal strength prediction, Planning of Future WiMAX Networks based on five developments (Advanced spatial techniques, mesh, relays and femtocells, self-configuring networks, cooperative WiMAX and cognitive) radios.

Wi-Fi Wireless Networking

Basic concept on Wireless Fidelity (Wi-Fi) network, Wi-Fi frequency bands and channels, Wi-Fi data rates, Deployment of Wi-Fi network with special emphasis on Planning and Design.

LEADERSHIP SKILLS

- **President**, International Student Association (ISA), University of Wyoming, 1000 E University Ave, Laramie, WY 82071, USA, *From May 2024 – May 2025*
- **Member**, Bangladesh Student Association (BSA), University of Wyoming, 1000 E University Ave, Laramie, WY 82071, USA, *From August 2023 – Present*
- **Former President**, Thakurgaon District Student Association (TZS), University of Rajshahi, Rajshahi - 6205, Bangladesh, *From 2014 – 2016*
- **Former General Secretary**, Lahiri Student Welfare Association (LSKP), Lahiri, Thakurgaon - 5041, Bangladesh, *From 2013 – 2014*
- **Former Organizing Secretary**, Rajshahi University Career Club, University of Rajshahi, Rajshahi - 6205, Bangladesh, *From May 2013 – June 2014*

I do hereby declare that the information given is true and correct to the best of my knowledge.

-Md Omor Faruk